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# SDM Institute of Technology, Ujire
# Department of Information Science and Engineering
# Major-Project Synopsis
# 6th Semester ISE-2026

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| **Title** | Design and Implementation of a Cloud-First Thin-Client Smart Keyboard Workstation |
| **Abstract** | This project proposes a cloud-first smart keyboard workstation for portable web development and automation. A Raspberry Pi operates as a thin client with a built-in mini projector for compact visual output. All development and computational tasks are handled through a cloud-hosted environment with AI-assisted automation. The system provides a low-cost, low-power, and scalable solution for student and field-based use. |
| **Objectives** | ● To design a compact smart keyboard workstation using a Raspberry Pi and mini projector.<br>● To enable cloud-based development through a thin-client interface.<br>● To integrate AI-assisted agents for automated coding and task management.<br>● To develop a low-cost, low-power, and scalable portable development system. |
| **Methodology** | The system is built using a Raspberry Pi configured with a lightweight Linux operating system and connected to a compact mini projector for display. A browser-based thin-client interface is used to access a cloud-hosted development environment over the internet. AI-assisted automation agents are deployed in the cloud to perform coding, testing, and task execution. The entire setup is optimized to minimize power consumption, hardware cost, and local processing load. |
| **Proposed outcome** | The proposed system delivers a functional portable workstation for web development using minimal local hardware. It enables users to access powerful cloud-based development tools through a compact smart keyboard interface. AI-assisted automation improves productivity by handling repetitive coding and task management operations. The design significantly reduces power consumption and hardware cost compared to traditional laptops. The solution demonstrates a scalable and practical model for cloud-first development environments. |
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